Body Control Module – High-Level Design

AUTOSAR Classic Platform | Sample Reference Architecture Document

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Prepared for internal engineering review and pilot evaluation

AUTOSAR Classic Platform | High-Level Design Document

Document Control

Field	Details
Document Title	Body Control Module – High-Level Design
Version	1.0
Document Type	Software Architecture / High-Level Design (HLD) Document
Purpose	Describe the software component architecture, interfaces, signals, and dependencies of the Body Control Module for engineering review and downstream integration.
Intended Audience	System architects, software developers, integration engineers, test engineers
Status	Sample / Reference (Synthetic Data)
Confidentiality	Internal use (sample/synthetic data for pilot evaluation)

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Introduction and Scope

This document describes the High-Level Design (HLD) of the Body Control Module (BCM) software, developed on the AUTOSAR Classic Platform. The BCM is responsible for managing body electronics functions including central locking, exterior lighting, wiper control, and window lift supervision. This document is intended to support architecture review, integration planning, and traceability to safety and test artifacts.

Scope covers the following software components:

- SWC_DoorLockControl
- SWC_ExteriorLightControl
- SWC_WiperControl
- SWC_WindowLiftSupervisor

Software Component Overview

The BCM software is decomposed into four Application Software Components (SWCs), each mapped to a dedicated runnable set and scheduled via the RTE. The table below summarizes each component's responsibility and its associated ECU.

Component Name	Responsibility	Mapped ECU
SWC_DoorLockControl	Central locking, door status monitoring, anti-theft interlock	BCM_ECU_01
SWC_ExteriorLightControl	Headlamp, tail lamp, and turn indicator control logic	BCM_ECU_01
SWC_WiperControl	Wiper speed control, rain sensor integration	BCM_ECU_01
SWC_WindowLiftSupervisor	Window motor control, anti-pinch protection	BCM_ECU_01

Ports and Interfaces

Each SWC exposes Sender-Receiver (S/R) and Client-Server (C/S) ports as defined below. Port interfaces are used by the RTE to route signals between components and the Basic Software (BSW) stack.

Port Name	Direction	Interface Type	Owning Component
PPort_DoorLockCmd	Provided	Sender-Receiver	SWC_DoorLockControl
RPort_DoorStatus	Required	Sender-Receiver	SWC_DoorLockControl
PPort_LampState	Provided	Sender-Receiver	SWC_ExteriorLightControl
RPort_AmbientLightSensor	Required	Sender-Receiver	SWC_ExteriorLightControl
PPort_WiperSpeedCmd	Provided	Sender-Receiver	SWC_WiperControl
RPort_RainSensorData	Required	Sender-Receiver	SWC_WiperControl
PPort_WindowPosition	Provided	Sender-Receiver	SWC_WindowLiftSupervisor
CSPort_DiagnosticService	Required	Client-Server	SWC_WindowLiftSupervisor

Signals and Data Elements

The following signals are exchanged across the ports described above. Data types conform to the AUTOSAR platform type definitions used in this project.

Signal Name	Data Type	Range / Values	Description
DoorLockCmd	uint8	0 = Unlock, 1 = Lock	Command to actuate central locking
DoorStatus	uint8	0 = Closed, 1 = Open	Current status of each door contact
LampState	uint8	0 = Off, 1 = Low, 2 = High	Exterior lamp state
AmbientLightLevel	uint16	0–1023 (ADC counts)	Ambient light sensor reading
WiperSpeedCmd	uint8	0–3 (Off/Low/Med/High)	Commanded wiper speed
RainIntensity	uint8	0–255	Rain sensor intensity value
WindowPosition	uint16	0–4095	Window position (potentiometer counts)
PinchDetected	boolean	TRUE / FALSE	Anti-pinch obstruction flag

Component Dependencies

The following dependencies exist between BCM software components and shared Basic Software modules.

Component	Depends On	Dependency Type
SWC_DoorLockControl	BSW_DIO, BSW_NvM	Digital I/O read/write, non-volatile state storage
SWC_ExteriorLightControl	SWC_DoorLockControl	Welcome-lighting trigger on unlock event
SWC_WiperControl	BSW_ADC	Rain sensor analog input
SWC_WindowLiftSupervisor	BSW_PWM, BSW_DiagnosticEventManager	Motor drive control, fault reporting

Functional Flow – Central Locking Sequence

The following sequence describes the door lock command flow from driver input to actuator response:

- 1. Driver presses the lock button; the HMI switch signal is captured by BSW_DIO.
- 2. SWC_DoorLockControl receives the DoorLockCmd signal via its provided port.
- 3. The component validates vehicle speed and door-ajar status before actuation.
- 4. On successful validation, the lock actuator relay is driven via BSW_DIO output.
- 5. DoorStatus is updated and broadcast to SWC_ExteriorLightControl for welcome lighting.
- 6. The final lock state is persisted using BSW_NvM for power-cycle recovery.

Assumptions and Open Points

- Assumes a single BCM ECU variant; multi-ECU BCM topologies are out of scope for this sample.
- Anti-pinch force threshold calibration values are placeholders and require validation.
- Diagnostic Trouble Code (DTC) mapping for window lift faults is not yet finalized.

Revision History

Version	Date	Description
0.1	August 2026	Initial draft – component identification
1.0	September 2026	Added ports, signals, dependencies, and functional flow